User

Yes let's try this. Use principles from general relativity if it helps

ChatGPT

Yes. There is a much cleaner derivation than the one we were using before. For this **single-step radiative transition**, the operator

$$\hat{a}\,\hat{b}^{\dagger}$$

does not need to be inserted by hand. It follows from interpreting your F as the **quantized electric force** and then quantizing both the oscillator coordinate and the radiation field.

This is also gauge-theoretically sound: the familiar electric-dipole interaction can be derived from gauge-invariant nonrelativistic QED, rather than postulated independently. (Phys. Rev. Research 2, 013206)

Start with your equation

Your equation is

$$\boxed{i\hbar\frac{\partial\Psi_1}{\partial t}+\frac{\hbar^2}{m}\frac{\partial^2\Psi_1}{\partial x^2}+i\hbar\frac{\partial\Psi_2}{\partial t}+\widehat{F}\,\hat{x}\,\Psi_2=0.} \tag{1}$$

The question is:

> Can we determine the structure of $\widehat{F}\hat{x}$ from physics rather than choosing $ab^{\dagger}+a^{\dagger}b$ ourselves?

For electromagnetic radiation, the answer is yes.

The natural identification is $F = qE$

For a charged oscillator,

$$\boxed{\widehat{F}=q\,\widehat{E}.} \tag{2}$$

This is just the electric part of the Lorentz force.

Notice what your equation then does. Move its Fx term to the right:

$$i\hbar\frac{\partial\Psi_2}{\partial t}=-\left[i\hbar\frac{\partial\Psi_1}{\partial t}+\frac{\hbar^2}{m}\frac{\partial^2\Psi_1}{\partial x^2}\right]-q\widehat{E}\,\hat{x}\Psi_2. \tag{3}$$

Therefore the interaction appearing in the Ψ_2 evolution is

$$\boxed{\widehat{H}_{\text{int}}=-q\hat{x}\,\widehat{E}.} \tag{4}$$

But this is exactly the electric-dipole interaction,

$$H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}, \qquad \mathbf{d} = q\mathbf{x}.$$

MIT's quantum-optics treatments use precisely this dipole interaction and, after quantizing the field, obtain matter and photon raising/lowering operators. (MIT Ultrafast Optics notes)

So your sign convention is actually rather convenient:

$$+\widehat{F}\widehat{x} \quad \text{on the left}$$

becomes

$$-\widehat{F}\widehat{x}$$

in the Ψ_2 evolution, giving the usual negative dipole interaction.

Now quantize x—we haven't introduced emission yet

For a harmonic oscillator,

$$\boxed{\widehat{x} = x_0 \left(\widehat{a} + \widehat{a}^\dagger\right),} \tag{5}$$

where

$$\boxed{x_0 = \sqrt{\frac{\hbar}{2m\omega_0}}.}$$

This follows simply from canonical quantization of the oscillator.

Nothing about photons has been assumed.

Quantize the electric force

A single electromagnetic mode has a quantized electric field of the form

$$\boxed{\widehat{E} = \mathcal{E}_0 \left(\widehat{b} + \widehat{b}^\dagger\right)} \tag{6}$$

up to an irrelevant phase convention.

For an electromagnetic mode of frequency ω_γ,

$$\mathcal{E}_0 \sim \sqrt{\frac{\hbar\omega_\gamma}{2\epsilon_0 \mathcal{V}}},$$

with mode-shape and polarization factors included when needed.

Therefore

$$\widehat{F} = q\mathcal{E}_0(\widehat{b} + \widehat{b}^\dagger).$$

Now multiply by $\widehat{x}$:

$$\widehat{F}\widehat{x} = q\mathcal{E}_0 x_0(\widehat{a} + \widehat{a}^\dagger)(\widehat{b} + \widehat{b}^\dagger).$$

Define

$$\boxed{\hbar g = q\mathcal{E}_0 x_0.} \tag{7}$$

Then

$$\boxed{\widehat{F}\widehat{x} = \hbar g(\widehat{a} + \widehat{a}^\dagger)(\widehat{b} + \widehat{b}^\dagger).} \tag{8}$$

And now expand it:

$$\boxed{\widehat{F}\widehat{x} = \hbar g\left[\widehat{a}\widehat{b} + \widehat{a}\widehat{b}^\dagger + \widehat{a}^\dagger\widehat{b} + \widehat{a}^\dagger\widehat{b}^\dagger\right].} \tag{9}$$

This is the result we wanted.

We did **not** insert

$$ab^\dagger$$

by hand.

It emerged because

$$\boxed{x \sim a + a^\dagger}$$

and

$$\boxed{F = qE \sim b + b^\dagger.}$$

Now apply it to your one-step emission

Take

$$\boxed{|\Psi_1\rangle = |1\rangle_{\rm osc}|0\rangle_\gamma.} \tag{10}$$

The desired emitted state is

$$\boxed{|f\rangle = |0\rangle_{\rm osc}|1\rangle_\gamma.} \tag{11}$$

Calculate the matrix element directly:

$$\langle 0,1|\widehat{F}\widehat{x}|1,0\rangle.$$

Using Eq. (9),

$$\langle 0,1|\widehat{F}\widehat{x}|1,0\rangle = \hbar g\langle 0,1|[ab + ab^\dagger + a^\dagger b + a^\dagger b^\dagger]|1,0\rangle.$$

Consider the four terms.

The first gives

$$b|0\rangle = 0.$$

The second gives

$$ab^\dagger|1,0\rangle = |0,1\rangle.$$

The third again contains

$$b|0\rangle = 0.$$

The fourth gives

$$a^\dagger b^\dagger|1,0\rangle \propto |2,1\rangle,$$

which is orthogonal to $|0,1\rangle$.

Therefore

$$\boxed{\langle 0,1|\widehat{F}\widehat{x}|1,0\rangle = \hbar g.} \tag{12}$$

The **only term contributing to this one-step emission matrix element is**

$$\boxed{\widehat{a}\,\widehat{b}^\dagger.}$$

That is a strong result.

For this particular matrix element, we did not even need to invoke the rotating-wave approximation.

The state selection itself picks out

$$ab^\dagger.$$

The reverse transition also emerges

Consider

$$|0,1\rangle \to |1,0\rangle.$$

Now

$$\boxed{\langle 1,0|Fx|0,1\rangle = \hbar g,}$$

and the surviving operator is

$$\boxed{a^\dagger b.}$$

Thus the pair

$$\boxed{ab^\dagger + a^\dagger b} \tag{13}$$

isn't something we needed to guess.

It is precisely the part of the full

$$(Fx) = \hbar g(a + a^\dagger)(b + b^\dagger)$$

operator that connects the resonant one-excitation states.

MIT's atomic-physics treatment describes exactly this structure: quantizing both the atomic dipole and electric field produces four operator products, while the resonant terms correspond to losing one matter excitation and creating one photon, or the reverse process. (MIT Atomic and Optical Physics II notes)

Energy conservation selects the same term independently

There is a second route to the same conclusion.

In the interaction picture,

$$a(t) = a\, e^{-i\omega_0 t},$$

and

$$b(t) = b\, e^{-i\omega_\gamma t}.$$

Therefore the four pieces of Eq. (9) evolve as

$$ab \sim e^{-i(\omega_0+\omega_\gamma)t},$$

$$ab^\dagger \sim e^{-i(\omega_0-\omega_\gamma)t},$$

$$a^\dagger b \sim e^{+i(\omega_0-\omega_\gamma)t},$$

and

$$a^\dagger b^\dagger \sim e^{+i(\omega_0+\omega_\gamma)t}.$$

For emission near resonance,

$$\omega_\gamma \simeq \omega_0.$$

Thus

$$ab^\dagger$$

varies slowly, whereas

$$ab$$

and

$$a^\dagger b^\dagger$$

oscillate at approximately

$$2\omega_0.$$

Averaging over times long compared with $1/\omega_0$ removes the fast terms.

So

$$\boxed{\widehat{F}\widehat{x} \longrightarrow \hbar g\left(ab^\dagger + a^\dagger b\right).} \tag{14}$$

This is the rotating-wave approximation, but here it emerges *after* deriving the complete Fx operator. (MIT Atomic and Optical Physics II, RWA notes)

The photon energy also emerges without putting it into Fx

The oscillator levels are

$$E_n = \hbar\omega_0\left(n + \frac{1}{2}\right).$$

Therefore

$$E_1 - E_0 = \hbar\omega_0.$$

For

$$|1,0\rangle \to |0,1_\gamma\rangle,$$

energy conservation requires

$$\hbar\omega_\gamma = \hbar\omega_0,$$

so

$$\boxed{\omega_\gamma = \omega_0.} \tag{15}$$

Thus the two key quantities have different origins:

$$\boxed{E_\gamma = \hbar\omega_0}$$

comes from energy conservation and the oscillator level spacing, while

$$\boxed{g = \frac{qx_0\mathcal{E}_0}{\hbar}}$$

comes from the electric-force coupling strength.

So the framework does not confuse transition energy with transition rate.

Put this back into your equation

Now your equation becomes

$$\boxed{i\hbar\frac{\partial\Psi_1}{\partial t} + \frac{\hbar^2}{m}\frac{\partial^2\Psi_1}{\partial x^2} + i\hbar\frac{\partial\Psi_2}{\partial t} + q\widehat{E}\widehat{x}\,\Psi_2 = 0.} \tag{16}$$

Equivalently,

$$\boxed{i\hbar\frac{\partial\Psi_2}{\partial t} = -\left[i\hbar\frac{\partial\Psi_1}{\partial t} + \frac{\hbar^2}{m}\frac{\partial^2\Psi_1}{\partial x^2}\right] - q\widehat{E}\widehat{x}\,\Psi_2.} \tag{17}$$

The last term is

$$\boxed{-q\widehat{E}\widehat{x} = -\hbar g(a + a^\dagger)(b + b^\dagger).}$$

And for the transition

$$|1,0\rangle \to |0,1\rangle,$$

its relevant matrix element is

$$\boxed{-\hbar g\, ab^{\dagger}.} \tag{18}$$

So for this very simple process, the ladder structure of Fx is no longer arbitrary.

Something else interesting happens with the Ψ_1 terms

Project your entire equation onto the final state

$$\langle 0, 1|.$$

The first part gives

$$\left\langle 0, 1 \left| i\hbar\dot{\Psi}_1 + \frac{\hbar^2}{m}\partial_x^2\Psi_1 \right\rangle .\right.$$

But

$$\Psi_1 = |1, 0\rangle.$$

Neither the harmonic-oscillator kinetic term nor its time evolution creates a photon.

Therefore

$$\boxed{\langle 0, 1| \left[i\hbar\dot{\Psi}_1 + \frac{\hbar^2}{m}\partial_x^2\Psi_1 \right] = 0.} \tag{19}$$

So **in the actual emitted-photon channel**, your equation reduces to

$$\boxed{i\hbar\langle 0, 1|\dot{\Psi}_2\rangle + \langle 0, 1|Fx|\Psi_2\rangle = 0.} \tag{20}$$

That is a surprisingly clean result.

The original Ψ_1 terms do not directly populate the emitted-photon state.

The transition into that state occurs through

$$\boxed{Fx.}$$

That is much closer to the physical role you've been assigning to the Fx term.

What does GR contribute?

General relativity gives a useful **structural analogy**, but it does not improve the photon derivation.

In weak-field GR, matter couples to the metric perturbation schematically through

$$\boxed{H_{\text{int}}^{\text{grav}} \sim -\frac{1}{2}\int h_{\mu\nu}T^{\mu\nu}\,d^3x.} \tag{21}$$

The pattern is the same:

$$\boxed{\text{matter operator} \times \text{field operator.}}$$

Quantize

$$h_{\mu\nu}$$

and it becomes a combination of graviton creation and annihilation operators, just as

$$E$$

became photon creation and annihilation operators.

So GR supports the conceptual principle that an interaction term naturally becomes a **product of operators acting on two quantum subsystems**.

However, gravitational radiation is quadrupolar rather than electric-dipolar. For a harmonic oscillator this generally leads to structures involving x^2 and hence two-quantum/squeezing-type transitions, rather than the simple

$$\Delta n = \pm 1$$

electric-dipole process. Quantum analyses of linearized gravitational-wave coupling to harmonic oscillators indeed emphasize its quadrupolar/squeezing character. (arXiv:hep-ph/9407410)

This gives a nice comparison:

electromagnetism	$q\,xE$	$\Delta n = \pm 1$,
gravity	quadrupole $\sim x^2 h$	Δn typically even.

That distinction actually supports rather than undermines the derivation.

Where this leaves your equation

For this simple single-step electromagnetic emission, we can now get much further than before.

We did **not** assume

$$Fx = \hbar g(ab^\dagger + a^\dagger b).$$

Instead:

$$\boxed{F = qE}$$

comes from the electromagnetic force,

then

$$\boxed{x = x_0(a + a^\dagger)}$$

comes from quantizing the oscillator,

and

$$\boxed{E = \mathcal{E}_0(b + b^\dagger)}$$

comes from quantizing radiation.

Therefore

$$\boxed{Fx = \hbar g(a + a^\dagger)(b + b^\dagger).}$$

The particular one-step transition itself selects

$$\boxed{ab^\dagger}$$

for emission and

$$\boxed{a^\dagger b}$$

for absorption.

That means the **operator structure of the simple emission term can indeed be derived from physical principles rather than inserted by hand.**

The remaining open question for your theory is narrower now: your equation still does not itself tell us that F must be the quantized electromagnetic force qE, nor does it derive Maxwell/QED from its bare form. But **once F is given its literal physical meaning as force, electromagnetic gauge theory fixes $F = qE$, and the one-photon ladder coupling follows automatically.**

That is a materially stronger result than our earlier construction.